Notes on
Advanced Mathematical Biology (AMB) IV

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NB: Notes for Chapter 5, Chapter 6.1, Chapter 7.1 and Chapter 7.2 from the Advanced Mathematical Biology IV module.

Contents

Definitions A–Z

Apparent viscosity

Axisymmetric

Body force

Continuum hypothesis

Flow behaviour index

Flow curve

Fluid consistency coefficient

Fluid density

Fluid velocity

Fully developed flow

Incompressible flow

Infinite shear viscosity

Laminar flow

Linear strain rate

Mass conservation equation

Mass flux

Mean fluid density

Momentum conservation law

Navier–Stokes equation

Newtonian fluid

No-slip boundary condition

Power-law

Pressure-driven flow

Regularity condition

Shear strain rate tensor

Shear-thinning fluids

Steady flow

Surface force

Time-dependent non-Newtonian fluids

Time-independent non-Newtonian fluids

Total mass flow rate

Total volume flow rate

Unsteady flow

Viscosity

Viscoelastic fluids

Volume-average velocity

Zero shear viscosity

1 Introduction to Viscous Fluids

1.1 Continuum Hypothesis

Key Points:

- Fluids treated as a continuos media.

The continuum hypothesis states that fluids, liquids or gases can be regarded as a continuos media whose properties change in a continuos manner - this is valid for $\lambda \ll \ell$ where λ is the mean-free path and ℓ is the length scale.

We have two properties of interest: fluid density $\rho(\mathbf{x}, t)$ and fluid velocity $\mathbf{u}(\mathbf{x}, t)$.

The mean fluid density is the mass per unit volume, denoted as $\bar{\rho}$.

In cases when the fluid is not homogeneous, then $\rho(\mathbf{x}, t) \neq \bar{\rho}$.

We further define,

$$\rho(\mathbf{x}, t) = \lim_{\delta V \to \delta V_0} \frac{\delta m}{\delta V}$$

where δV_0 is the limiting volume - that is the smallest volume with homogeneous mass.

1.2 Fluid Motion Descriptions

Key Points:

- Eulerian description: fixed frame.
- Lagrangian description: moving frame.

We have two different descriptions of fluid as it moves through a volume V.

Eulerian description: (fixed frame) In this description, we observe a fluid velocity $\mathbf{u}(\mathbf{x}, t)$ at a certain point $\mathbf{x}$ where the observer is fixed.

Lagrangian description: (moving frame) In this description, we follow a particle of fluid starting at $\mathbf{x}_0$ and observe a fluid velocity $\tilde{\mathbf{u}}(\mathbf{x}_0, t)$.

1.3 Time Derivatives

Key Points:

- Local derivative $\frac{\partial}{\partial t}$.
- Material derivative
$$\frac{\mathrm{D}}{\mathrm{D}t} = \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla.$$

Let $f(\mathbf{x}, t)$ be a fluid property.

In the Eulerian description,
$$\frac{\mathrm{d}f}{\mathrm{d}t} = \frac{\partial f}{\partial t}\frac{\mathrm{d}t}{\mathrm{d}t} = \frac{\partial f}{\partial t}$$
where $\frac{\partial}{\partial t}$ is the local derivative at a fixed point.

In the Lagrangian description,
$$\begin{aligned}\frac{\mathrm{d}f}{\mathrm{d}t} &= \frac{\partial f}{\partial t}\frac{\mathrm{d}t}{\mathrm{d}t} + \frac{\partial f}{\partial x} \cdot \frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f}{\partial y} \cdot \frac{\mathrm{d}y}{\mathrm{d}t} + \frac{\partial f}{\partial z} \cdot \frac{\mathrm{d}z}{\mathrm{d}t} \\ &= \frac{\partial f}{\partial t} + \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} \cdot \nabla f = \underbrace{\left[\frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla\right]}_{\frac{\mathrm{D}}{\mathrm{D}t}} f\end{aligned}$$

where $\frac{\mathrm{D}}{\mathrm{D}t} := \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla$ is the material derivative.

1.4 Volume Flow Rate, Mass Flow Rate and Mass Flux

Key Points:

- Total volume flow rate
$$Q = \int_s \mathbf{u} \cdot \hat{\mathbf{n}} \, \mathrm{d}S.$$
- Total mass flow rate
$$\dot{m} = \int_S \rho(\mathbf{u} \cdot \hat{\mathbf{n}}) \, \mathrm{d}S$$
- Mass flux $= \rho(\mathbf{u} \cdot \hat{\mathbf{n}})$.

Considering a fluid mass m enclosed by a volume V with surface S.

Following the derivation, we have the key points:

$$\mathrm{d}x = \mathbf{u} \cdot \hat{\mathbf{n}} \, \mathrm{d}t \quad \text{and} \quad \mathrm{d}V' = \mathbf{u} \cdot \hat{\mathbf{n}} \, \mathrm{d}t \, \mathrm{dS}$$

where V' is the volume of the traced out parallelepiped and $\hat{\mathbf{n}}$ the unit normal to the surface.

We end with an expression for the total volume flow rate through S,

$$Q = \int_S \mathbf{u} \cdot \hat{\mathbf{n}} \, \mathrm{d}S$$

Again, we have the key points from the derivation:

$$\mathrm{d}m' = \rho \, \mathrm{d}V' = \rho \mathbf{u} \cdot \hat{\mathbf{n}} \, \mathrm{d}t \, \mathrm{d}S$$

where m' is the mass of fluid passing through S in a time t and we are assuming a homogeneous distribution of mass inside $\mathrm{d}V'$, so ρ is constant inside $\mathrm{d}V'$.

This gives an expression for the total mass flow rate through S,

$$\dot{m} = \int_S \rho(\mathbf{u} \cdot \hat{\mathbf{n}}) \, \mathrm{dS}$$

Further, if m is homogeneous throughout all of S, then ρ is constant and we have

$$\dot{m} = \rho Q$$

The object $\mathbf{u} \cdot \hat{\mathbf{n}}$ describes the (in/out)flow of the fluid if it is positive/negative with the case $\mathbf{u} \cdot \hat{\mathbf{n}} = 0$ describing the case that the fluid flows parallel to the surface.

The mass flux is the fluid mass flow across a unit area per unit time,

$$\text{mass flux} = \frac{\dot{m}}{\mathrm{d}S} = \rho \mathbf{u} \cdot \hat{\mathbf{n}}$$

1.5 Mass Conservation

Key Points:

- Mass conservation/continuity equation

$$\frac{\mathrm{D}\rho}{\mathrm{D}t} + \rho(\nabla \cdot \mathbf{u}) = 0.$$

- Steady flow: $\rho = \rho(\mathbf{x})$.
- Unsteady flow: $\rho = \rho(\mathbf{x}, t)$.
- Incompressible flow: $\frac{\mathrm{D}\rho}{\mathrm{D}t} = 0$ which results in the conservation law reading

$$\nabla \cdot \mathbf{u} = 0.$$

Consider again a fluid mass m enclosed by a volume V with surface S.

The law can be written as,

$$\text{Rate of decrease of mass inside } V = \text{Net rate at which mass leaves } V \text{ through } S$$

Using,

$$\text{Rate of decrease of mass inside } V = -\frac{\mathrm{d}m}{\mathrm{d}t} = -\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho \,\mathrm{d}V$$

$$\text{Net rate at which mass leaves } V \text{ through } S = \dot{m} = \int_S \rho(\mathbf{u}\cdot\hat{\mathbf{n}})\,\mathrm{dS}$$

and differentiation rules so find the integral form of the mass conservation equation (or continuity equation) as,

$$\int_V \frac{\partial \rho}{\partial t}\,\mathrm{d}V + \int_S \rho(\mathbf{u}\cdot\hat{\mathbf{n}})\,\mathrm{d}S = 0$$

which we can write in differential form using the divergence theorem and vector calculus identities as

$$\frac{\partial \rho}{\partial t} + (\nabla\rho)\cdot\mathbf{u} + \rho(\nabla\cdot\mathbf{u}) = 0 \iff \frac{\mathrm{D}\rho}{\mathrm{D}t} + \rho(\nabla\cdot\mathbf{u}) = 0$$

1.5.1 Steady Flow

Steady flow is when $\rho = \rho(\mathbf{x})$, that the density is time independent and unsteady flow is when $\rho = \rho(\mathbf{x}, t)$.

Under the steady flow assumption, the conservation law reads,

$$\int_S \rho(\mathbf{u}\cdot\hat{\mathbf{n}})\,\mathrm{dS} = 0$$

that is, $|\dot{m}_{\mathrm{in}}| = |\dot{m}_{\mathrm{out}}|$ and the mass flows entering must equal the mass flow leaving the volume V.

1.5.2 Incompressible Flow

Incompressible is when the density does not change as the fluid moves through space, that is $\frac{\mathrm{D}\rho}{\mathrm{D}t} = 0$.

Under the incompressible assumption, the conservation law reads,

$$\nabla\cdot\mathbf{u} = 0$$

and using the divergence theorem, this is equivalent to,

$$\int_S \mathbf{u}\cdot\hat{\mathbf{n}}\,\mathrm{d}S = 0$$

that is, $|Q_{\mathrm{in}}| = |Q_{\mathrm{out}}|$ and the total volume flow entering must equal the total volume flow leaving the volume V.

We also have the volume-average velocity,

$$\langle\mathbf{u}\rangle = \frac{1}{S}\int_S \mathbf{u}\cdot\hat{\mathbf{n}}\,\mathrm{d}S$$

1.6 Momentum Conservation

Key Points:

- Momentum conservation law

$$\rho\frac{\mathrm{D}\mathbf{u}}{\mathrm{D}t} = -\nabla p + \nabla\cdot\overline{\overline{\tau}} + \rho\mathbf{g}.$$

Again, consider a fluid mass m enclosed by a volume V with surface S. We have two different forces acting on the fluid.

Body force: These are forces acting on every fluid particle as it moves through V, for instance gravity.

Surface force: These are forces acting on the fluid particles adjacent to the surface S.

We will consider the surface forces p and $\overline{\overline{\tau}}$, which are pressure and stress.

We will view pressure as the normal force per unit area due to a change in momentum. It is compressive. that is it points in the $-\hat{\mathbf{n}}$ direction.

We will view the stress as the force per unit area due to particles of fluid exerting upon each other.

Stress has a normal component, and a tangential component, denoted as normal stress and shear stress.

Let $\mathbf{g}$ be the net body force per unit mass inside V. We have the following forces:

$$\begin{aligned}\mathbf{F}_{\text{body}} &= \int_V \rho\mathbf{g}\,\mathrm{d}V\\ \mathbf{F}_{\text{pres}} &= -\int_S p\hat{\mathbf{n}}\,\mathrm{d}S\\ \mathbf{F}_{\text{str}} &= \int_S \overline{\overline{\tau}}\hat{\mathbf{n}}\,\mathrm{d}S\end{aligned}$$

which are the total body, pressure and stress forces. These give rise to the total force,

$$\mathbf{F} = \int_V \rho\mathbf{g}\,\mathrm{d}V - \int_S p\hat{\mathbf{n}}\,\mathrm{d}S + \int_S \overline{\overline{\tau}}\hat{\mathbf{n}}\,\mathrm{d}S$$

The time rate of change of momentum has two components,

(i) The net momentum flowing out of V.

(ii) The time rate of change of momentum due to unsteady fluctuations of the fluid properties inside V.

That is,

$$\frac{\mathrm{d}}{\mathrm{d}t}(m\mathbf{u}) = \int_S \rho(\mathbf{u}\cdot\hat{\mathbf{n}})\mathbf{u}\,\mathrm{d}S + \frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho\mathbf{u}\,\mathrm{d}V$$

and using Newton's second law we arrive at the integral form of the momentum conservation law,

$$\int_V \rho\mathbf{g}\,\mathrm{d}V - \int_S p\hat{\mathbf{n}}\,\mathrm{d}S + \int_S \overline{\overline{\tau}}\hat{\mathbf{n}}\,\mathrm{d}S = \int_S \rho(\mathbf{u}\cdot\hat{\mathbf{n}})\mathbf{u}\,\mathrm{d}S + \frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho\mathbf{u}\,\mathrm{d}V$$

We can then use the following:

$$-\int_S p\hat{\mathbf{n}}\,\mathrm{d}S = -\int_V \nabla p\,\mathrm{d}V, \quad \int_S \overline{\overline{\tau}}\cdot\hat{\mathbf{n}}\,\mathrm{d}S = \int_V \nabla\cdot\overline{\overline{\tau}}\,\mathrm{d}V, \quad \frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho\mathbf{u}\,\mathrm{d}V = \int_V \frac{\partial(\rho\mathbf{u})}{\partial t}\,\mathrm{d}V$$

We wish to use the divergence theorem for the term

$$\int_S \rho(\mathbf{u}\cdot\hat{\mathbf{n}})\mathbf{u}\,\mathrm{d}S$$

yet the integrand is a <u>vector</u> not a scalar. In index notation,

$$\int_S \rho u_i n_i u_j\,\mathrm{d}S = \int_S T_{ij}n_i\,\mathrm{d}S$$

where we have defined the tensor $\overline{\overline{T}}$ with components $T_{ij} = \rho u_i u_j$. Now, we can use the divergence theorem,

$$\int_S T_{ij}n_i\,\mathrm{d}S = \int_V \frac{\partial T_{ij}}{\partial x_i}\,\mathrm{d}V = \int_V \nabla\cdot\overline{\overline{T}}\,\mathrm{d}V$$

In summary, we can use

$$\int_S \rho(\mathbf{u}\cdot\hat{\mathbf{n}})\mathbf{u}\,\mathrm{d}S = \int_V \nabla\cdot\overline{\overline{T}}\,\mathrm{d}V$$

and the other identities, to obtain,

$$\int_V \left[\frac{\partial(\rho\mathbf{u})}{\partial t} + \nabla\cdot\overline{\overline{T}} + \nabla p - \nabla\cdot\overline{\overline{\tau}} - \rho\mathbf{g}\right]\mathrm{d}V = 0$$

as our momentum conservation law. We then require,

$$\frac{\partial(\rho\mathbf{u})}{\partial t} + \nabla\cdot\overline{\overline{T}} = -\nabla p + \nabla\cdot\overline{\overline{\tau}} + \rho\mathbf{g}$$

with a little bit of work,

$$\begin{aligned}[\nabla\cdot\overline{\overline{T}}]_j = \frac{\partial T_{ij}}{\partial x_i} = \frac{\partial}{\partial x_i}(\rho u_i u_j) &= \frac{\partial}{\partial x_i}(\rho u_i)u_j + \rho u_i\frac{\partial u_j}{\partial x_i}\\ &= \frac{\partial}{\partial x_i}(\rho u_i)u_j + \rho u_i\frac{\partial}{\partial x_i}(u_j)\end{aligned}$$

which we can identify as

$$\nabla\cdot\overline{\overline{T}} = [\nabla\cdot(\rho\mathbf{u})]\mathbf{u} + \rho(\mathbf{u}\cdot\nabla)\mathbf{u}$$

Using the product rule, we obtain

$$\rho\left[\frac{\partial \mathbf{u}}{\partial t}+(\mathbf{u}\cdot\nabla)\mathbf{u}\right]+\underbrace{\left[\frac{\partial \rho}{\partial t}+\nabla\cdot(\rho\mathbf{u})\right]}_{\text{by conservation law }=0}\mathbf{u}=-\nabla p+\nabla\cdot\overline{\overline{\tau}}+\rho\mathbf{g}$$

which results in the differential form of the momentum conservation law,

$$\rho\left[\frac{\partial \mathbf{u}}{\partial t}+(\mathbf{u}\cdot\nabla)\mathbf{u}\right]=-\nabla p+\nabla\cdot\overline{\overline{\tau}}+\rho\mathbf{g}\iff\rho\frac{\mathrm{D}\mathbf{u}}{\mathrm{D}t}=-\nabla p+\nabla\cdot\overline{\overline{\tau}}+\rho\mathbf{g}$$

1.7 Motion and Deformation of Fluids

Key Points:

- Shear strain rate tensor

$$\dot{\gamma}_{ij}=\frac{\partial u_i}{\partial x_j}+\frac{\partial u_j}{\partial x_i}.$$

1.7.1 Motion of Fluids

We will consider four types of motion:

(i) Translation: Movement with shape and orientation unchanged.

(ii) Rotation: Movement with shape unchanged but orientation altered.

(iii) Linear strain: Under normal stress, orientation unchanged but shape altered

- Under positive normal stress, we will have tension.
- Under negative normal stress, we will have compression.

(iv) Shear strain: Both shape and orientation changed.

We denote the rate of change if linear/shear strain as the linear/shear strain rate .

1.7.2 Deformation of Fluids

The fluid's shape changes under two types of strain: linear and shear.

Linear strain: We have the linear strain rate e_{ii} where

$$e_{ii}=\frac{\partial u_i}{\partial x_i}$$

Shear strain: We have the shear strain rate tensor $\overline{\overline{\dot{\gamma}}}$ with components $\dot{\gamma}_{ij}$ where

$$\dot{\gamma}_{ij}=\frac{\partial u_i}{\partial x_j}+\frac{\partial u_j}{\partial x_i}$$

1.8 Newtonian Fluids

Key Points:

- Constitutive equation for an incompressible Newtonian fluid
$$\tau_{ij} = \mu\dot{\gamma}_{ij}.$$
- Navier-Stokes equation (NSE)
$$\rho\frac{\mathrm{D}\mathbf{u}}{\mathrm{D}t} = -\nabla p + \mu\nabla^2\mathbf{u} + \rho\mathbf{g}$$
.

Searching for the constitutive equation $\overline{\overline{\tau}} = f(\overline{\overline{\dot{\gamma}}})$ for the fluid, we first define a Newtonian fluid as a fluid in which the stress depends linearly on strain rates.

For an incompressible fluid,

$$\overline{\overline{\tau}} = \mu\overline{\overline{\dot{\gamma}}} \quad \text{or} \quad \tau_{ij} = \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$$

where μ is the viscosity.

Assuming a constant fluid viscosity,

$$\nabla\cdot\overline{\overline{\tau}} = \nabla\cdot(\mu\overline{\overline{\dot{\gamma}}}) = \mu\nabla\cdot\overline{\overline{\dot{\gamma}}}$$

and with a bit of work,

$$\begin{aligned}[\nabla\cdot\overline{\overline{\dot{\gamma}}}]_j = \frac{\partial\dot{\gamma}_{ij}}{\partial x_i} &= \frac{\partial}{\partial x_i}\left[\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right] \\ &= \frac{\partial}{\partial x_i}\left(\frac{\partial u_i}{\partial x_j}\right) + \frac{\partial^2 u_i}{\partial x_i^2}\end{aligned}$$

which we can identify with

$$\nabla\cdot\overline{\overline{\dot{\gamma}}} = \nabla(\nabla\cdot\mathbf{u}) + \nabla^2\mathbf{u} = \nabla^2\mathbf{u}$$

since our fluid is incompressible, which leaves us with the Navier-Stokes equation

$$\rho\left[\frac{\partial\mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u}\right] = -\nabla p + \mu\nabla^2\mathbf{u} + \rho\mathbf{g}$$

which is simply the momentum conservation law for incompressible, Newtonian fluids.

2 Analytical solutions of viscous fluid flows

2.1 Poiseuille Flow

Key Points:

- The assumptions are: (i) incompressible, (ii) Newtonian, (iii) Pressure driven, (iv) Steady flow, (v) Laminar flow, (vi) Axisymmetric flow, (vii) Fully developed flow and (viii) Gravity is negligible.
- Under these assumptions
$$u_x(r) = u_{\max}\left[1 - \left(\frac{r}{R}\right)^2\right].$$

Consider the simplest case of fluid flow, that is flow through a rigid cylindrical tube of radius R and length L.

Due to the cylindrical symmetry, we will use the cylindrical coordinates, labelled (r, θ, x).

We make the following assumptions:

(i) The fluid is incompressible.

(ii) The fluid is Newtonian.

(iii) The flow is pressure driven, that is the pressure varies only in the direction of flow (which we take as the x-direction).

(iv) The flow is steady.

(v) The flow is laminar, that is the fluid travels in smooth, parallel layers, without mixing. Thus, $\mathbf{u} = (u_r, u_\theta, u_x) = (0, 0, u_x)$.

(vi) The flow is axisymmetric, that is the fluid velocity is independent of the azimuthal, θ, direction due to symmetry.

(vii) The flow is fully developed, that is the fluid velocity, and hence shear stress, does not vary in the direction of flow, namely $\frac{\partial u_x}{\partial x} = 0$. As a result, $u_x = u_x(r)$.

(viii) The effect of gravity on the fluid is negligible.

The continuity equation (in cylindrical coordinates) reads,

$$\frac{\partial \rho}{\partial t} + \frac{1}{r}\frac{\partial}{\partial r}(r\rho u_r) + \frac{1}{r}\frac{\partial}{\partial \theta}(\rho u_\theta) + \frac{\partial}{\partial x}(\rho u_x) = 0$$

Using assumption (v),

$$\frac{1}{r}\frac{\partial}{\partial r}(r\rho u_r) = 0, \quad \frac{1}{r}\frac{\partial}{\partial \theta}(\rho u_\theta) = 0$$

and under assumption (iv),

$$\frac{\partial \rho}{\partial t} = 0$$

which recovers assumption (vii):

$$\frac{\partial u_x}{\partial x} = 0$$

Doing the same argument for the Navier-Stokes equations, the components read:

r-component:

$$\frac{\partial p}{\partial r} = 0$$

θ-component:

$$\frac{\partial p}{\partial \theta} = 0$$

x-component:

$$\frac{\partial p}{\partial x} = \frac{\mu}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_x}{\partial r}\right)$$

The r, θ- components recover assumption (iii).

Since $p = p(x)$ and $u_x = u_x(r)$, the x-component reads,

$$\frac{\mathrm{d}p}{\mathrm{d}x} = \frac{\mu}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}u_x}{\mathrm{d}r}\right)$$

which gives each side equal to a constant c_0 as the LHS is x dependent and the RHS is r dependent, that is

$$\frac{\mathrm{d}p}{\mathrm{d}x} = c_0 \quad \text{and} \quad \frac{\mu}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}u_x}{\mathrm{d}r}\right) = c_0$$

Using $p(0) = p_0$ and $p(L) = p_L$,

$$p(x) = p_0 - \frac{p_0 - p_L}{L}x = p_0 - \frac{\Delta p}{L}x$$

which now gives

$$\frac{\mu}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}u_x}{\mathrm{d}r}\right) = -\frac{\Delta p}{L}$$

which can be integrated to read

$$\frac{\mathrm{d}u_x}{\mathrm{d}r} = -\frac{\Delta p}{2\mu L}r + \frac{c_2}{r}$$

for some constant c_2. The regularity condition is the condition that $\left.\frac{\mathrm{d}u_x}{\mathrm{d}r}\right|_{r=0} = 0$; this is due to the shear stress at the centreline $r = 0$ being finite. Using this condition and integrating,

$$u_x(r) = -\frac{\Delta p}{4\mu L}r^2 + c_3$$

for some constant c_3. The no slip boundary condition is the condition $u_x(R) = u_x(-R) = 0$; this is due to the cylinder being modelled as rigid, and so the fluid velocity vanishes at the boundary. Using this, we arrive at,

$$u_x(r) = u_{\max}\left[1 - \left(\frac{r}{R}\right)^2\right]$$

where we have defined $u_{\max} = \frac{\Delta p R^2}{4\mu L}$.

3 Non-Newtonian Fluids

3.1 Introduction

Key Points:

- $\mu_{\text{app}} = \frac{\tau}{\dot{\gamma}}$.

The plot of shear stress against strain rate is known as the flow curve.

For non-Newtonian fluids, we define

$$\mu_{\text{app}} = \frac{\tau_{ij}}{\dot{\gamma}_{ij}}$$

as the apparent viscosity also written as $\mu_{\text{app}} = \tau/\dot{\gamma}$.

There are three general classes of fluids based on their rheology:

1. Time-independent non-Newtonian fluids (Generalised Newtonian fluids): These are fluids where the shear stress depends only on strain rate.
2. Time-dependent non-Newtonian fluids: These are fluids where the shear stress against strain rate depends on the duration of the shear.
3. Viscoelastic fluids: These are fluids that exhibit both fluid behaviour and elastic solid behaviour.

3.2 Time-independent non-Newtonian fluids

Key Points:

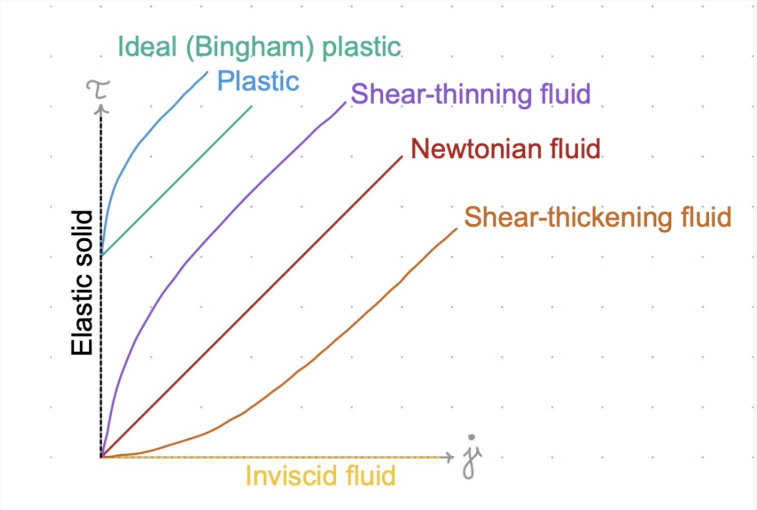
- Power law fluids $\tau = \kappa\dot{\gamma}^n$ with constitutive equation

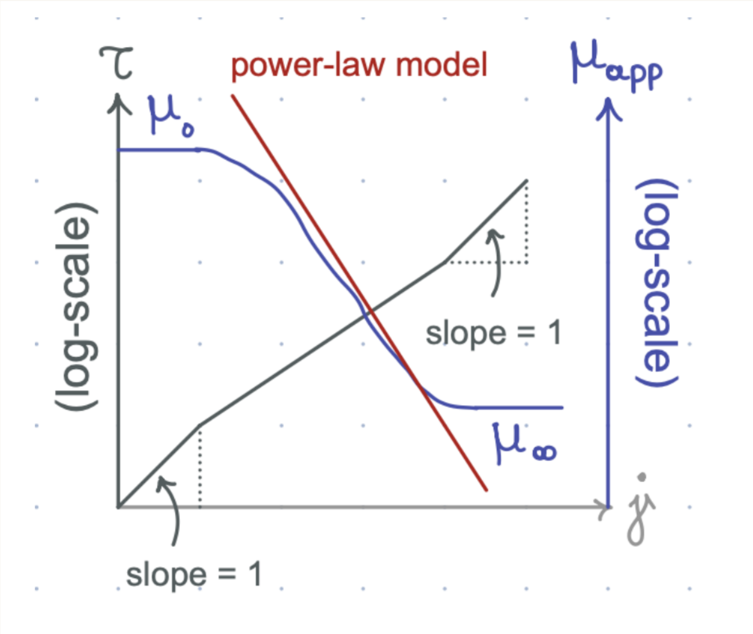
$$\tau = \kappa|\dot{\gamma}|^{n-1}\dot{\gamma}.$$

In simple shear, the constitutive equation is of the form $\tau = f(\dot{\gamma})$. Depending on the form of $f(\dot{\gamma})$ we have:

1. Shear-thinning fluids also known as pseudo-plastic fluids.
2. Shear-thickening fluids also known as dilatant fluids.
3. Plastic fluids.

Each of these types of fluids can be seen in the following diagram.





3.2.1 Shear-thinning fluids

These fluids have an apparent viscosity which decreases with increasing shear rate.

At very low/high strain rates, they exhibit Newtonian behaviour. The apparent viscosity at these low/high strain rates are μ_0 and μ_∞ which are the zero shear viscosity and infinite shear viscosity.

3.2.2 The power-law

The power-law state that

$$\tau = \kappa \dot{\gamma}^n$$

where κ is a scalar and known as the fluid consistency coefficient and also n is dimensionless and known as the flow behaviour index.

Moreover, if κ increases then the fluid becomes more viscous, if n decreases the fluid exhibits more non-Newtonian characteristics and $n \in [0, 1]$ with $n = 1$ being a Newtonian fluid.

Under this model, the apparent fluid viscosity is given by

$$\mu_{\text{app}} = \kappa|\dot{\gamma}|^{n-1}$$

and the constitutive equation is given by

$$\tau = \mu_{\text{app}}\dot{\gamma} = \kappa|\dot{\gamma}|^{n-1}\dot{\gamma}$$

This model is only applicable over a limited range of shear strain rate.

Example: Consider blood flowing through a rigid cylinder of radius R and length L. The fluid is modelled under the power-law and has the following assumptions:

(i) Incompressible.

(ii) Steady flow.

(iii) Laminar.

(iv) Axisymmetric.

(v) Negligible gravity.

Further, $p = p_0$ and $p = p_L$ at $x = 0$ and $x = L$.

1. Derive the velocity profile for blood flow inside vessels, under the above assumptions.
2. Given that blood is a shear-thinning fluid, make a qualitative plot of the normalised velocity with respect to the maximum velocity versus the normalised radial distance with respect to the vessel radius.

Solution: 1. Using cylindrical coordinates, the continuity equation reads,

$$\frac{\partial \rho}{\partial t} + \frac{1}{r}\frac{\partial}{\partial r}(r\rho u_r) + \frac{1}{r}\frac{\partial}{\partial \theta}(\rho u_\theta) + \frac{\partial}{\partial x}(\rho u_x) = 0$$

which under assumptions (ii) and (iii) reads,

$$\frac{\partial u_x}{\partial x} = 0$$

and we have the fully developed condition, hence $u_x = u_x(r)$.

Since the fluid is non-Newtonian, the components of the momentum conservation law read:
<u>r-component</u>:

$$\rho\left(\frac{\partial u_r}{\partial t} + u_r\frac{\partial u_r}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_r}{\partial \theta} - \frac{u_\theta^2}{r} + u_x\frac{\partial u_r}{\partial x}\right) = -\frac{\partial p}{\partial r} + \rho g_r$$
$$+ \left(\frac{1}{r}\frac{\partial(r\tau_{rr})}{\partial r} + \frac{1}{r}\frac{\partial \tau_{\theta r}}{\partial \theta} - \frac{\tau_{\theta\theta}}{r} + \frac{\partial \tau_{xr}}{\partial x}\right)$$

which under assumptions (iii) and (v) reads,

$$\frac{\partial p}{\partial r} = \frac{1}{r}\frac{\partial (r\tau_{rr})}{\partial r} + \frac{1}{r}\frac{\partial \tau_{\theta r}}{\partial \theta} - \frac{\tau_{\theta\theta}}{r} + \frac{\partial \tau_{xr}}{\partial x}$$

Using the power-law in index from

$$\tau_{ij} = \kappa \left| \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right|^{n-1} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

we can calculate the following values for the stress tensor,

$$\begin{aligned}
\tau_{rr} &= \kappa \left| \frac{\partial u_r}{\partial r} + \frac{\partial u_r}{\partial r} \right|^{n-1} \left(\frac{\partial u_r}{\partial r} + \frac{\partial u_r}{\partial r} \right) = 0 \\
\tau_{\theta r} &= \kappa \left| \frac{\partial u_\theta}{\partial r} + \frac{\partial u_r}{\partial \theta} \right|^{n-1} \left(\frac{\partial u_\theta}{\partial r} + \frac{\partial u_r}{\partial \theta} \right) = 0 \\
\tau_{xr} &= \kappa \left| \frac{\partial u_x}{\partial r} + \frac{\partial u_r}{\partial x} \right|^{n-1} \left(\frac{\partial u_r}{\partial x} + \frac{\partial u_x}{\partial x_r} \right) = \tau_{xr}(r)
\end{aligned}$$

which we can then use to arrive at

$$\frac{\partial p}{\partial r} = 0$$

<u>θ-component</u>:

$$\begin{aligned}
\rho \left(\frac{\partial u_\theta}{\partial t} + u_r \frac{\partial u_\theta}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_\theta}{\partial \theta} + \frac{u_r u_\theta}{r} + u_x \frac{\partial u_\theta}{\partial x} \right) &= -\frac{1}{r}\frac{\partial p}{\partial \theta} + \rho g_\theta \\
&\quad + \left(\frac{1}{r^2}\frac{\partial (r^2 \tau_{r\theta})}{\partial r} + \frac{1}{r}\frac{\partial \tau_{\theta\theta}}{\partial \theta} + \frac{\partial \tau_{x\theta}}{\partial x} \right)
\end{aligned}$$

which under assumptions (iii) and (v) reads,

$$\frac{1}{r}\frac{\partial p}{\partial \theta} = \frac{1}{r^2}\frac{\partial (r^2 \tau_{r\theta})}{\partial r} + \frac{1}{r}\frac{\partial \tau_{\theta\theta}}{\partial \theta} + \frac{\partial \tau_{x\theta}}{\partial x}$$

Using the power-law in index from

$$\tau_{ij} = \kappa \left| \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right|^{n-1} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

we can again calculate the following values for the stress tensor,

$$\begin{aligned}
\tau_{r\theta} &= \kappa \left| \frac{\partial u_r}{\partial \theta} + \frac{\partial u_\theta}{\partial r} \right|^{n-1} \left(\frac{\partial u_r}{\partial \theta} + \frac{\partial u_\theta}{\partial r} \right) = 0 \\
\tau_{\theta\theta} &= \kappa \left| \frac{\partial u_\theta}{\partial \theta} + \frac{\partial u_\theta}{\partial \theta} \right|^{n-1} \left(\frac{\partial u_\theta}{\partial \theta} + \frac{\partial u_\theta}{\partial \theta} \right) = 0 \\
\tau_{x\theta} &= \kappa \left| \frac{\partial u_x}{\partial \theta} + \frac{\partial u_\theta}{\partial x} \right|^{n-1} \left(\frac{\partial u_x}{\partial \theta} + \frac{\partial u_\theta}{\partial x} \right) = 0
\end{aligned}$$

where we used assumption (iv) for the final line. Applying this, we arrive at

$$\frac{\partial p}{\partial \theta} = 0$$

The r, θ-components of the momentum conservation law together lead to $p = p(r)$.

<u>x-component</u>:

$$\rho\left(\frac{\partial u_x}{\partial t} + u_r\frac{\partial u_x}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_x}{\partial \theta} + u_x\frac{\partial u_x}{\partial x}\right) = -\frac{\partial p}{\partial x} + \rho g_x + \left(\frac{1}{r}\frac{\partial (r\tau_{rx})}{\partial r} + \frac{1}{r}\frac{\partial \tau_{\theta x}}{\partial \theta} + \frac{\partial \tau_{xx}}{\partial x}\right)$$

which under assumptions (iii), (v) and time-independence reads,

$$\frac{\partial p}{\partial x} = \frac{1}{r}\frac{\partial (r\tau_{rx})}{\partial r} + \frac{1}{r}\frac{\partial \tau_{\theta x}}{\partial \theta} + \frac{\partial \tau_{xx}}{\partial x}$$

Using the power-law in index from

$$\tau_{ij} = \kappa\left|\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right|^{n-1}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$$

we can (for the last time) calculate the following values for the stress tensor,

$$\begin{aligned}
\tau_{rx} &= \kappa\left|\frac{\partial u_r}{\partial x} + \frac{\partial u_x}{\partial r}\right|^{n-1}\left(\frac{\partial u_r}{\partial x} + \frac{\partial u_x}{\partial r}\right) = \tau_{rx}(r)\\
\tau_{\theta x} &= \kappa\left|\frac{\partial u_\theta}{\partial x} + \frac{\partial u_x}{\partial \theta}\right|^{n-1}\left(\frac{\partial u_\theta}{\partial x} + \frac{\partial u_x}{\partial \theta}\right) = 0\\
\tau_{xx} &= \kappa\left|\frac{\partial u_x}{\partial x} + \frac{\partial u_x}{\partial x}\right|^{n-1}\left(\frac{\partial u_x}{\partial x} + \frac{\partial u_x}{\partial x}\right) = 0
\end{aligned}$$

where we used the fully developed condition for the final line. Consequently,

$$\frac{\partial p}{\partial x} = \frac{1}{r}\frac{\partial (r\tau_{rx})}{\partial r}$$

and since $p = p(x)$ and $\tau_{rx} = \tau_{rx}(r)$,

$$\frac{\mathrm{d}p}{\mathrm{d}x} = \frac{1}{r}\frac{\mathrm{d}(r\tau_{rx})}{\mathrm{d}r}$$

which requires both sides to be equal to a constant c_0,

$$\frac{\mathrm{d}p}{\mathrm{d}x} = c_0 \Rightarrow p(x) = c_0 x + c_1$$

for some c_1, and using our initial conditions,

$$p(x) = p_0 - \frac{\Delta p}{L}x$$

which further gives

$$\frac{1}{r}\frac{\mathrm{d}(r\tau_{rx})}{\mathrm{d}r} = -\frac{\Delta p}{L} \Rightarrow \tau_{rx}(r) = -\frac{\Delta p r}{2L} + \frac{c_2}{r}$$

for some c_2. As the shear stress must be finite along the centre line $r = 0$, we require $c_2 = 0$,

$$\tau_{rx}(r) = -\frac{\Delta p r}{2L}$$

which we can substitute into the power-law to give,

$$\kappa \left|\frac{\mathrm{d}u_x}{\mathrm{d}r}\right|^{n-1} \frac{\mathrm{d}u_x}{\mathrm{d}r} = -\frac{\Delta p r}{2L}$$

We can get rid of the absolute value by noticing that $\tau_{rx}(r) = -\frac{\Delta p r}{2L}$ implies that τ_{rx}, and consequently $\frac{\mathrm{d}u_x}{\mathrm{d}r}$, is negative in the positive r direction, that is $-|\frac{\mathrm{d}u_x}{\mathrm{d}r}| = \frac{\mathrm{d}u_x}{\mathrm{d}r}$. Hence,

$$-\kappa \left|\frac{\mathrm{d}u_x}{\mathrm{d}r}\right|^{n} = -\frac{\Delta p r}{2L} \Rightarrow \left|\frac{\mathrm{d}u_x}{\mathrm{d}r}\right| = \left(\frac{\Delta p r}{2\kappa L}\right)^{1/n} \Rightarrow \frac{\mathrm{d}u_x}{\mathrm{d}r} = -\left(\frac{\Delta p r}{2\kappa L}\right)^{1/n}$$

which we can integrate to get

$$u_x(r) = -\left(\frac{\Delta p}{2\kappa L}\right)^{1/n} \frac{r^{1+1/n}}{1+1/n} + c_3$$

for some c_3. Finally, applying the no slip condition due to the cylinder being modelled as rigid, we obtain the velocity profile

$$\boxed{u_x(r) = \left(\frac{\Delta p}{2\kappa L}\right)^{1/n} \frac{R^{1+1/n}}{1+1/n}\left[1 - \left(\frac{r}{R}\right)^{n-1}\right]}$$

2. Since blood is shear thinning we would plot $u_x(r)/u_{\max}$ against r/R can would obtain: